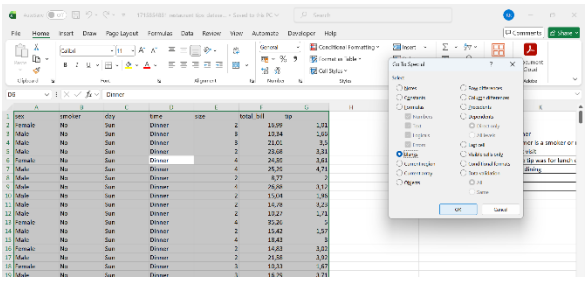
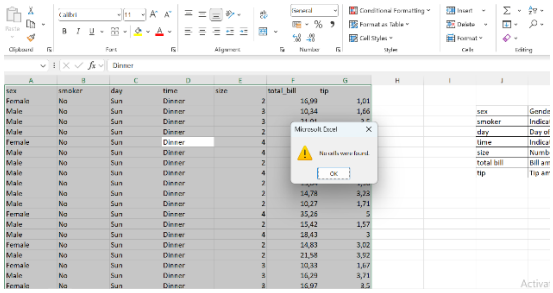
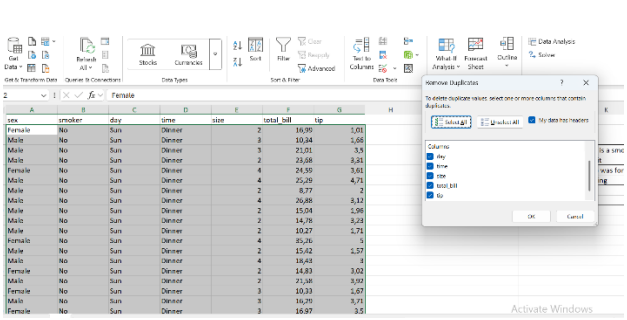
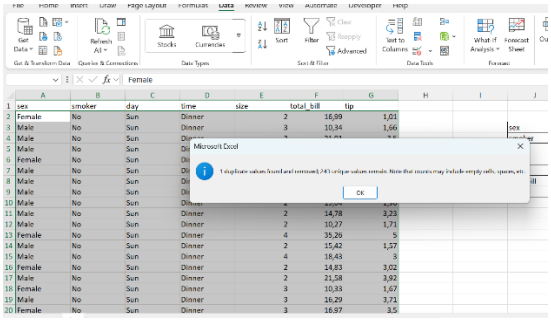
The document outlines a series of tasks related to statistical analysis, including identifying independent and dependent variables, constructing a multiple linear regression model, and examining correlations. It also references error metrics such as squared error, mean squared error, and root mean squared error, suggesting a focus on model evaluation and accuracy assessment .

Task2



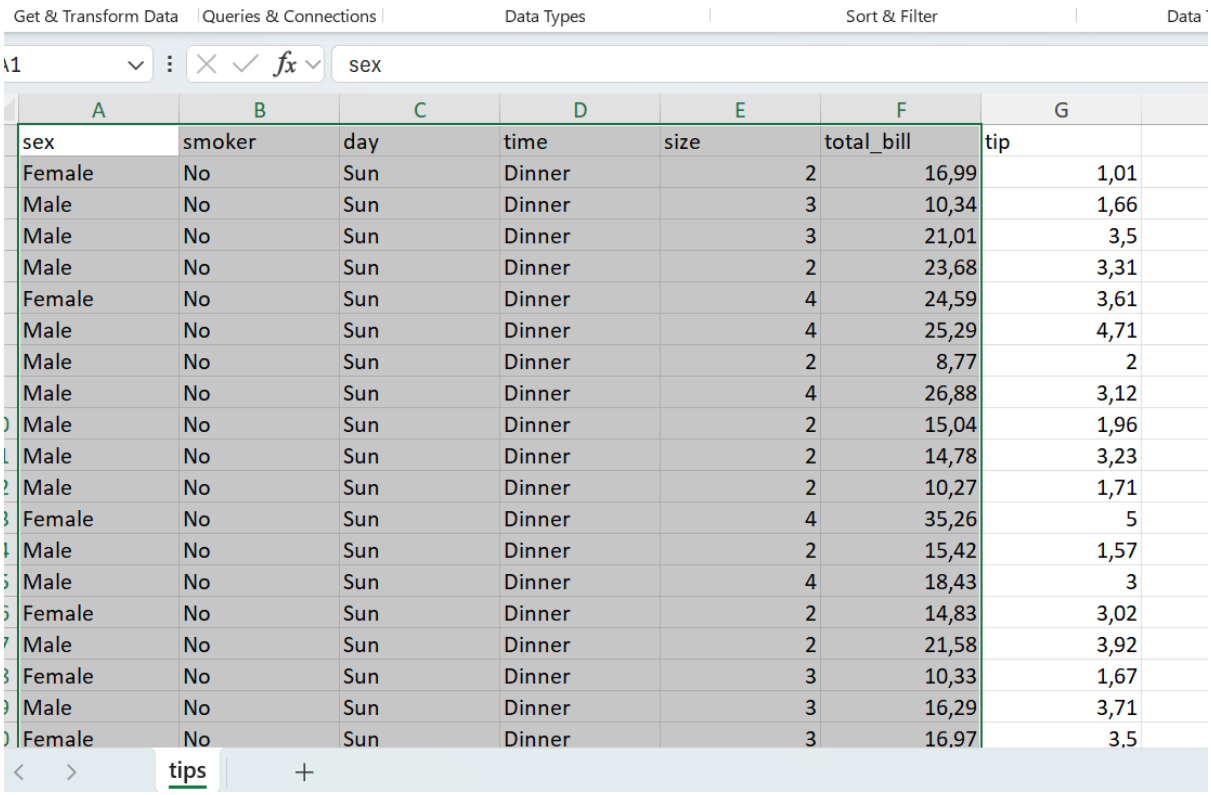






Task3

Independent variables



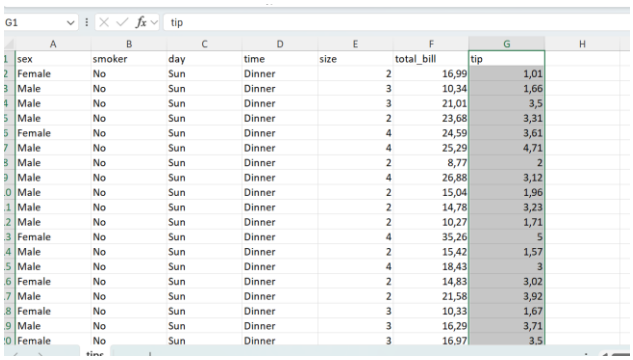
Get & Transform Data | Queries & Connections | Data Types | Sort & Filter | Data

A1 | sex

	A	B	C	D	E	F	G
	sex	smoker	day	time	size	total_bill	tip
	Female	No	Sun	Dinner	2	16,99	1,01
	Male	No	Sun	Dinner	3	10,34	1,66
	Male	No	Sun	Dinner	3	21,01	3,5
	Male	No	Sun	Dinner	2	23,68	3,31
	Female	No	Sun	Dinner	4	24,59	3,61
	Male	No	Sun	Dinner	4	25,29	4,71
	Male	No	Sun	Dinner	2	8,77	2
	Male	No	Sun	Dinner	4	26,88	3,12
	Male	No	Sun	Dinner	2	15,04	1,96
	Male	No	Sun	Dinner	2	14,78	3,23
	Male	No	Sun	Dinner	2	10,27	1,71
	Female	No	Sun	Dinner	4	35,26	5
	Male	No	Sun	Dinner	2	15,42	1,57
	Male	No	Sun	Dinner	4	18,43	3
	Female	No	Sun	Dinner	2	14,83	3,02
	Male	No	Sun	Dinner	2	21,58	3,92
	Female	No	Sun	Dinner	3	10,33	1,67
	Male	No	Sun	Dinner	3	16,29	3,71
	Female	No	Sun	Dinner	3	16,97	3,5

tips

dependent variables



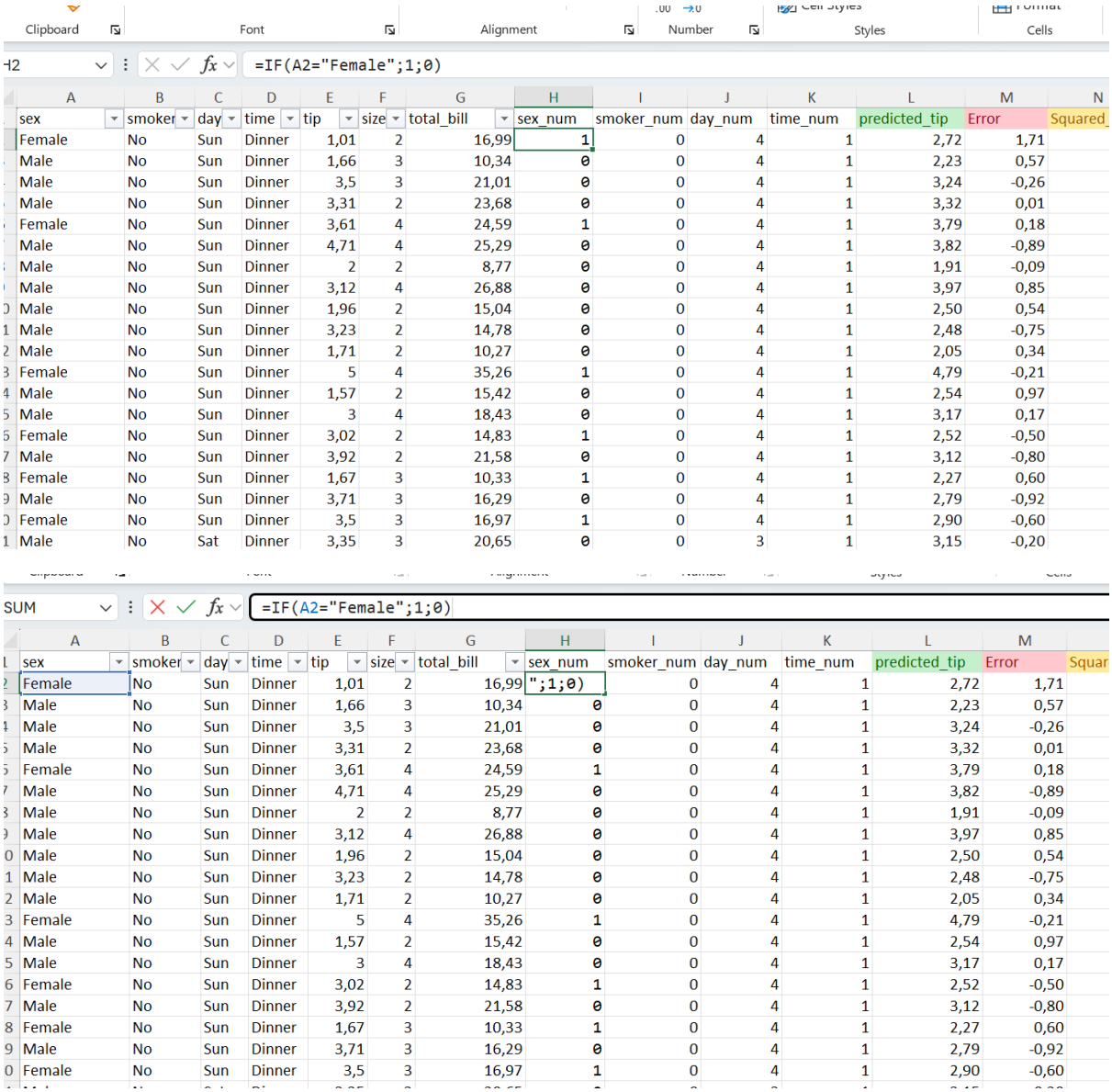
G1 | tip

	A	B	C	D	E	F	G	H
1	sex	smoker	day	time	size	total_bill	tip	
2	Female	No	Sun	Dinner	2	16,99	1,01	
3	Male	No	Sun	Dinner	3	10,34	1,66	
4	Male	No	Sun	Dinner	3	21,01	3,5	
5	Male	No	Sun	Dinner	2	23,68	3,31	
6	Female	No	Sun	Dinner	4	24,59	3,61	
7	Male	No	Sun	Dinner	4	25,29	4,71	
8	Male	No	Sun	Dinner	2	8,77	2	
9	Male	No	Sun	Dinner	4	26,88	3,12	
10	Male	No	Sun	Dinner	2	15,04	1,96	
11	Male	No	Sun	Dinner	2	14,78	3,23	
12	Male	No	Sun	Dinner	2	10,27	1,71	
13	Female	No	Sun	Dinner	4	35,26	5	
14	Male	No	Sun	Dinner	2	15,42	1,57	
15	Male	No	Sun	Dinner	4	18,43	3	
16	Female	No	Sun	Dinner	2	14,83	3,02	
17	Male	No	Sun	Dinner	2	21,58	3,92	
18	Female	No	Sun	Dinner	3	10,33	1,67	
19	Male	No	Sun	Dinner	3	16,29	3,71	
20	Female	No	Sun	Dinner	3	16,97	3,5	

Task4

Multiple linear regression model

Task 5



Clipboard | Font | Alignment | Number | Styles | Cells

H2 =IF(A2="Female";1;0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_
	Female	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	
	Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	
	Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	
	Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	
	Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	
	Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	
	Male	No	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	
	Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	
0	Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	
1	Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	
2	Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	
3	Female	No	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	
4	Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	
5	Male	No	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	
6	Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	
7	Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	
8	Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60	
9	Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	1	2,79	-0,92	
0	Female	No	Sun	Dinner	3,5	3	16,97	1	0	4	1	2,90	-0,60	
1	Male	No	Sat	Dinner	3,35	3	20,65	0	0	3	1	3,15	-0,20	

SUM =IF(A2="Female";1;0)

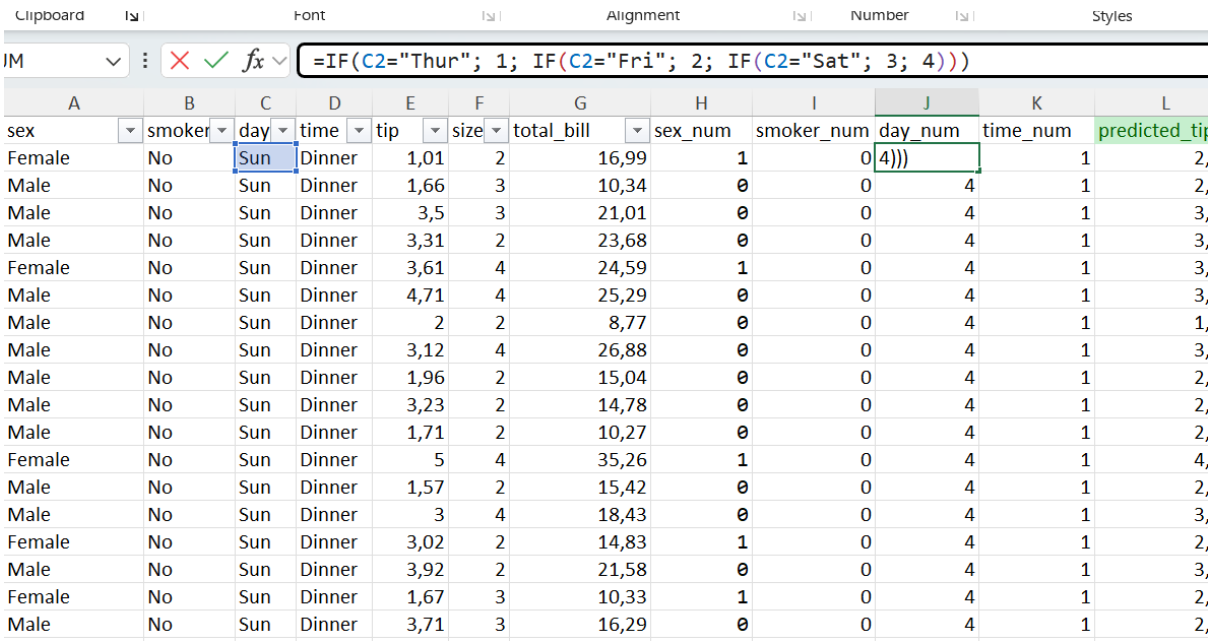
	A	B	C	D	E	F	G	H	I	J	K	L	M
l	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squar
2	Female	No	Sun	Dinner	1,01	2	16,99	";1;0)	0	4	1	2,72	1,71	
3	Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	
4	Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	
5	Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	
5	Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	
7	Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	
8	Male	No	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	
9	Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	
0	Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	
1	Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	
2	Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	
3	Female	No	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	
4	Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	
5	Male	No	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	
6	Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	
7	Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	
8	Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60	
9	Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	1	2,79	-0,92	
0	Female	No	Sun	Dinner	3,5	3	16,97	1	0	4	1	2,90	-0,60	

I2 =IF(B2="yes";1;0)

	A	B	C	D	E	F	G	H	I	J
1	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num
2	Female	No	Sun	Dinner	1,01	2	16,99	1	0	
3	Male	No	Sun	Dinner	1,66	3	10,34	0	0	
4	Male	No	Sun	Dinner	3,5	3	21,01	0	0	
5	Male	No	Sun	Dinner	3,31	2	23,68	0	0	
6	Female	No	Sun	Dinner	3,61	4	24,59	1	0	
7	Male	No	Sun	Dinner	4,71	4	25,29	0	0	
8	Male	No	Sun	Dinner	2	2	8,77	0	0	
9	Male	No	Sun	Dinner	3,12	4	26,88	0	0	
10	Male	No	Sun	Dinner	1,96	2	15,04	0	0	
11	Male	No	Sun	Dinner	3,23	2	14,78	0	0	
12	Male	No	Sun	Dinner	1,71	2	10,27	0	0	
13	Female	No	Sun	Dinner	5	4	35,26	1	0	
14	Male	No	Sun	Dinner	1,57	2	15,42	0	0	
15	Male	No	Sun	Dinner	3	4	18,43	0	0	
16	Female	No	Sun	Dinner	3,02	2	14,83	1	0	
17	Male	No	Sun	Dinner	3,92	2	21,58	0	0	
18	Female	No	Sun	Dinner	1,67	3	10,33	1	0	
19	Male	No	Sun	Dinner	3,71	3	16,29	0	0	
20	Female	No	Sun	Dinner	3,5	3	16,97	1	0	
21	Male	No	Sat	Dinner	3,35	3	20,65	0	0	

J2 =IF(C2="Thur"; 1; IF(C2="Fri"; 2; IF(C2="Sat"; 3; 4)))

	A	B	C	D	E	F	G	H	I	J	K
1	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_n
2	Female	No	Sun	Dinner	1,01	2	16,99	1	0	4	
3	Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	
4	Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	
5	Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	
6	Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	
7	Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	
8	Male	No	Sun	Dinner	2	2	8,77	0	0	4	
9	Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	
10	Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	
11	Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	
12	Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	
13	Female	No	Sun	Dinner	5	4	35,26	1	0	4	
14	Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	
15	Male	No	Sun	Dinner	3	4	18,43	0	0	4	
16	Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	
17	Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	
18	Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	
19	Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	
20	Female	No	Sun	Dinner	3,5	3	16,97	1	0	4	
21	Male	No	Sat	Dinner	3,35	3	20,65	0	0	3	

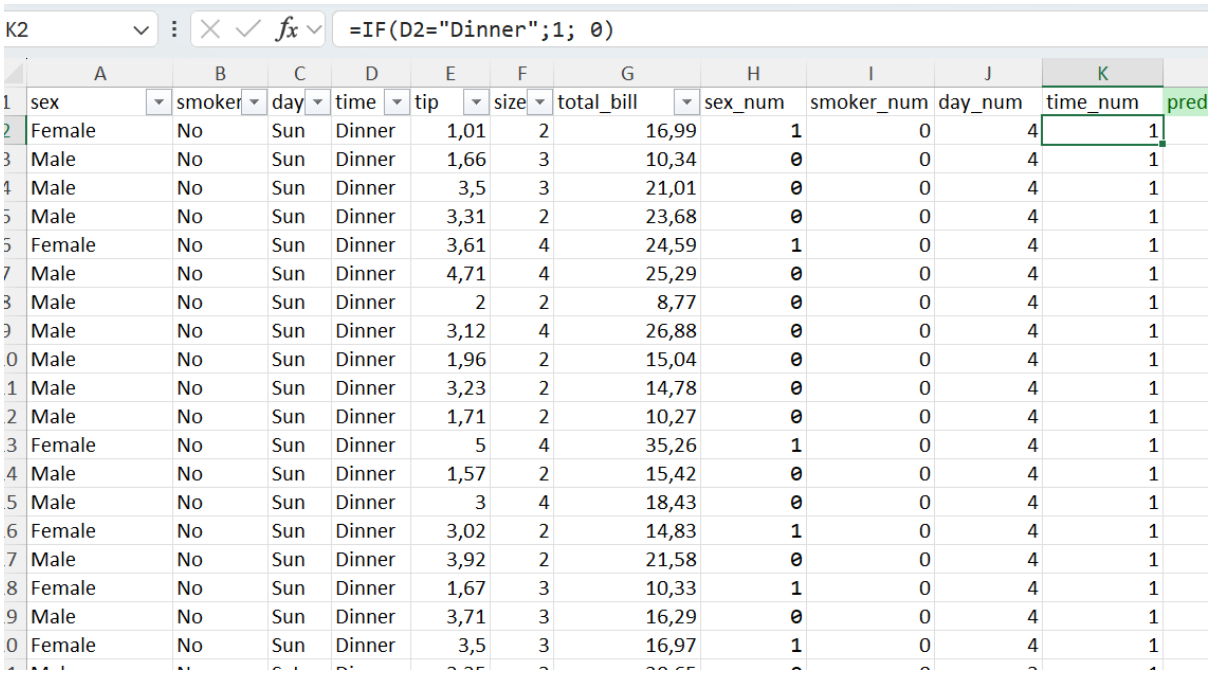


Clipboard | Font | Alignment | Number | Styles

IM

=IF(C2="Thur"; 1; IF(C2="Fri"; 2; IF(C2="Sat"; 3; 4)))

	A	B	C	D	E	F	G	H	I	J	K	L
	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip
	Female	No	Sun	Dinner	1,01	2	16,99	1	0	4)))	1	2,
	Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,
	Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,
	Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,
	Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,
	Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,
	Male	No	Sun	Dinner	2	2	8,77	0	0	4	1	1,
	Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,
	Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,
	Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,
	Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,
	Female	No	Sun	Dinner	5	4	35,26	1	0	4	1	4,
	Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,
	Male	No	Sun	Dinner	3	4	18,43	0	0	4	1	3,
	Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,
	Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,
	Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,
	Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	1	2,

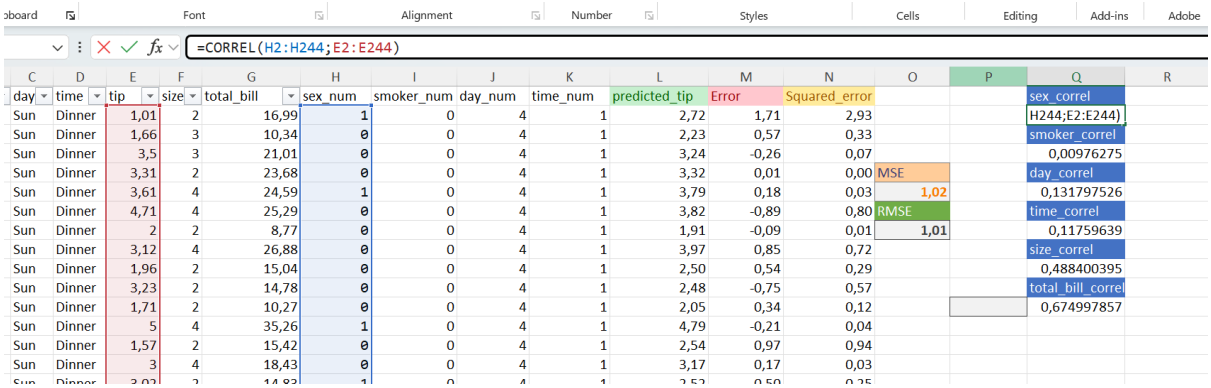


K2

=IF(D2="Dinner";1; 0)

	A	B	C	D	E	F	G	H	I	J	K	
1	sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	pred
2	Female	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	
3	Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	
4	Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	
5	Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	1	
6	Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	1	
7	Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	1	
8	Male	No	Sun	Dinner	2	2	8,77	0	0	4	1	
9	Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	1	
10	Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	1	
11	Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	1	
12	Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	1	
13	Female	No	Sun	Dinner	5	4	35,26	1	0	4	1	
14	Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	1	
15	Male	No	Sun	Dinner	3	4	18,43	0	0	4	1	
16	Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	1	
17	Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	1	
18	Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	1	
19	Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	1	
20	Female	No	Sun	Dinner	3,5	3	16,97	1	0	4	1	

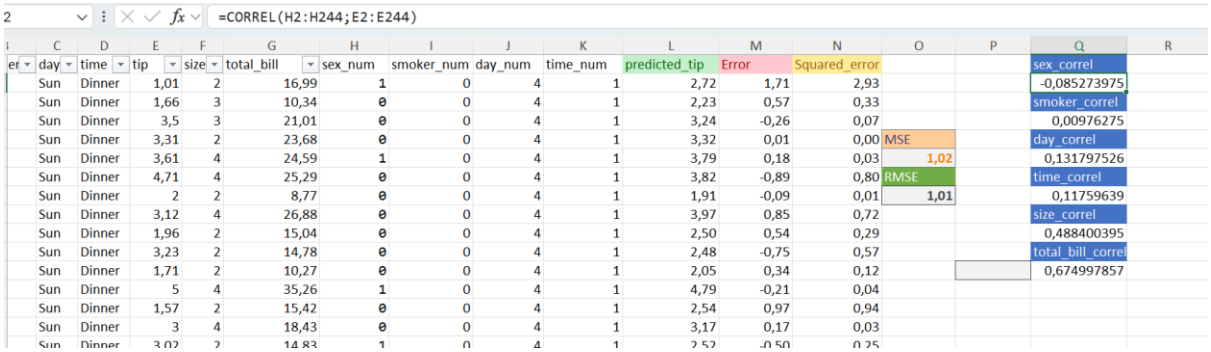
Correlations



day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error
Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93
Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33
Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07
Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00
Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03
Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80
Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01
Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72
Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29
Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57
Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12
Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04
Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94
Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03

MSE	1,02
RMSE	1,01

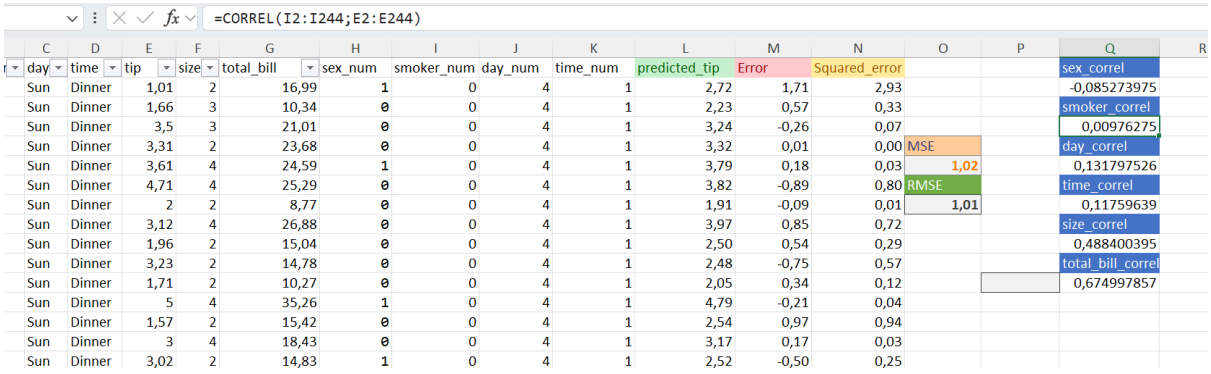
sex_correl
H244;E2:E244)
smoker_correl
0,00976275
day_correl
0,131797526
time_correl
0,11759639
size_correl
0,488400395
total_bill_correl
0,674997857



day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error
Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93
Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33
Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07
Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00
Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03
Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80
Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01
Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72
Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29
Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57
Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12
Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04
Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94
Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03

MSE	1,02
RMSE	1,01

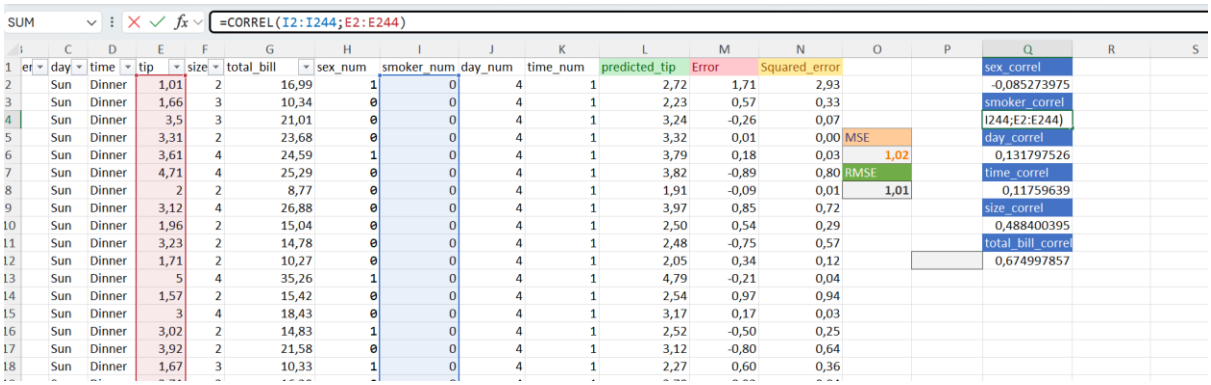
sex_correl
-0,085273975
smoker_correl
0,00976275
day_correl
0,131797526
time_correl
0,11759639
size_correl
0,488400395
total_bill_correl
0,674997857



day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error
Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93
Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33
Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07
Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00
Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03
Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80
Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01
Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72
Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29
Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57
Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12
Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04
Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94
Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03
Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25

MSE	1,02
RMSE	1,01

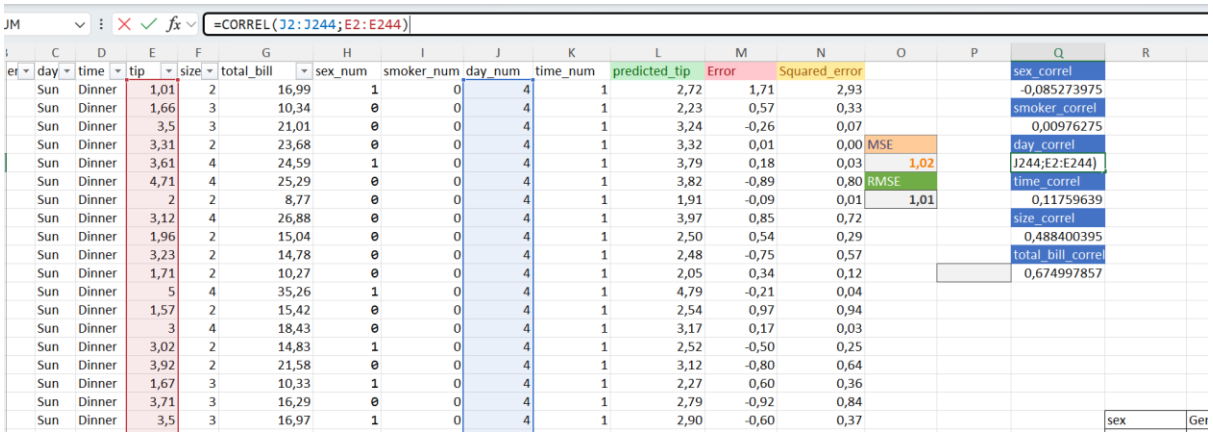
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smoker_correl
0,00976275
day_correl
0,131797526
time_correl
0,11759639
size_correl
0,488400395
total_bill_correl
0,674997857



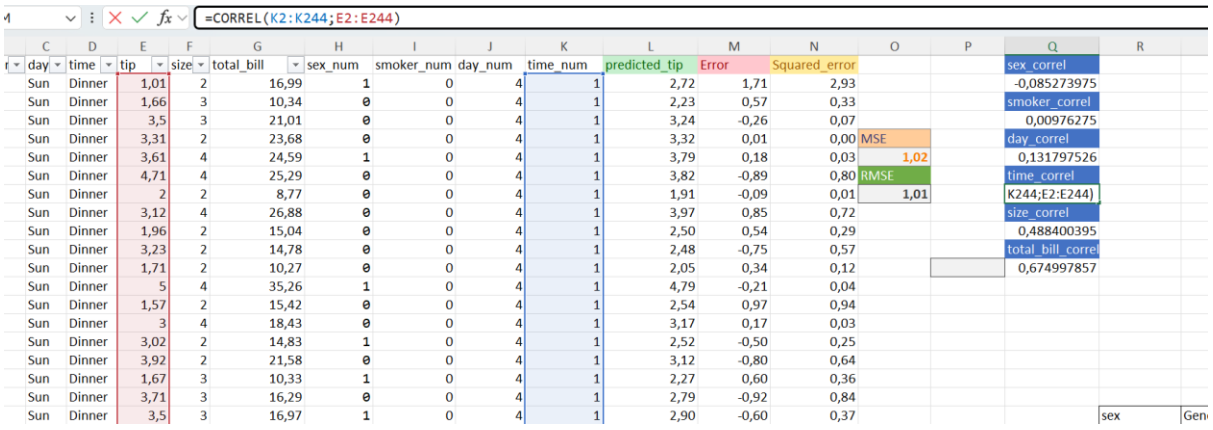
day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error
Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93
Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33
Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07
Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00
Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03
Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80
Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01
Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72
Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29
Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57
Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12
Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04
Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94
Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03
Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25
Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	0,64
Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60	0,36

MSE	1,02
RMSE	1,01

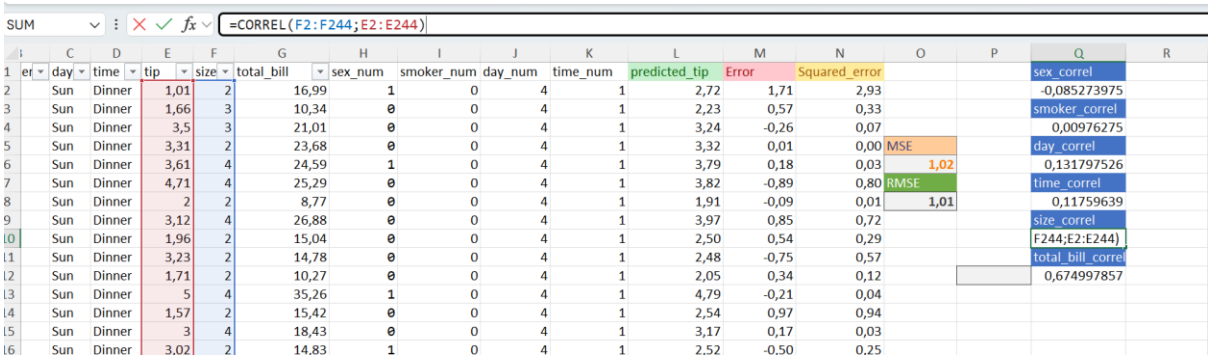
sex_correl
-0,085273975
smoker_correl
I244;E2:E244)
day_correl
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time_correl
0,11759639
size_correl
0,488400395
total_bill_correl
0,674997857



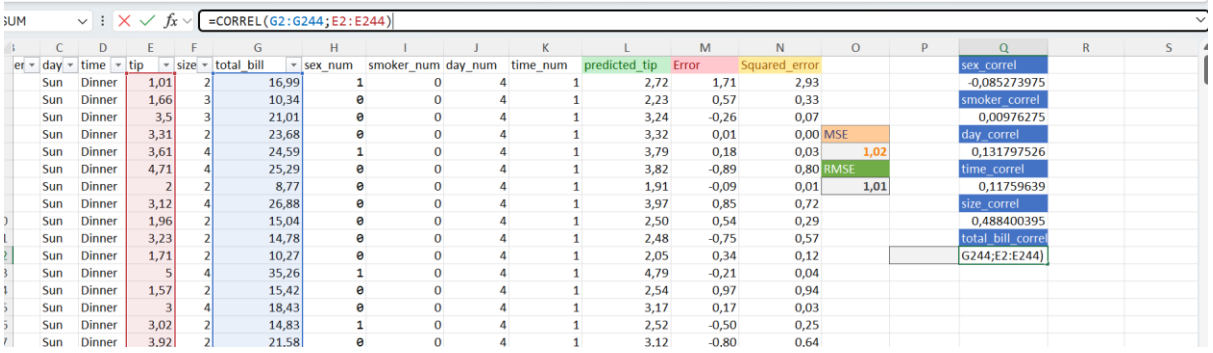
	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
er	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error			sex_correl	
	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93			-0,085273975	
	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33			smoker_correl	
	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07			0,00976275	
	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00	MSE		day_correl	
	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03	1,02		J244;E2:E244)	
	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80	RMSE		time_correl	
	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01	1,01		0,11759639	
	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72			size_correl	
	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29			0,488400395	
	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57			total_bill_correl	
	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12			0,674997857	
	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04				
	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94				
	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03				
	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25				
	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	0,64				
	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60	0,36				
	Sun	Dinner	3,71	3	16,29	0	0	4	1	2,79	-0,92	0,84				
	Sun	Dinner	3,5	3	16,97	1	0	4	1	2,90	-0,60	0,37				sex



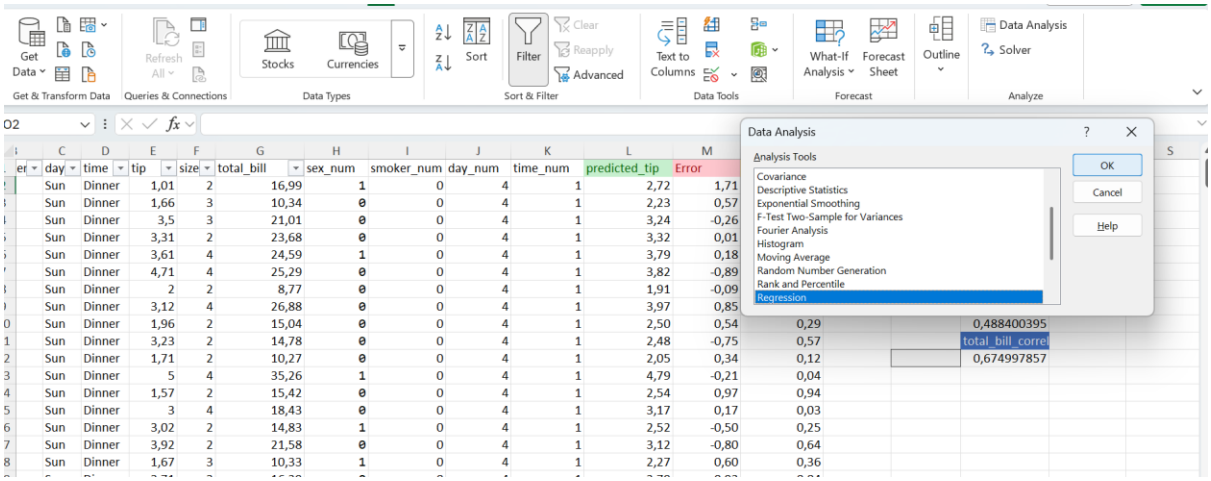
	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error			sex_correl	
	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93			-0,085273975	
	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33			smoker_correl	
	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07			0,00976275	
	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00	MSE		day_correl	
	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03	1,02		0,131797526	
	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80	RMSE		time_correl	
	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01	1,01		K244;E2:E244)	
	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72			size_correl	
	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29			0,488400395	
	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57			total_bill_correl	
	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12			0,674997857	
	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04				
	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94				
	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03				
	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25				
	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	0,64				
	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60	0,36				
	Sun	Dinner	3,71	3	16,29	0	0	4	1	2,79	-0,92	0,84				
	Sun	Dinner	3,5	3	16,97	1	0	4	1	2,90	-0,60	0,37				sex



	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
er	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error			sex_correl	
	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93			-0,085273975	
	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33			smoker_correl	
	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07			0,00976275	
	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00	MSE		day_correl	
	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03	1,02		0,131797526	
	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80	RMSE		time_correl	
	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01	1,01		0,11759639	
	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72			size_correl	
	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29			F244;E2:E244)	
	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57			total_bill_correl	
	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12			0,674997857	
	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04				
	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94				
	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03				
	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25				

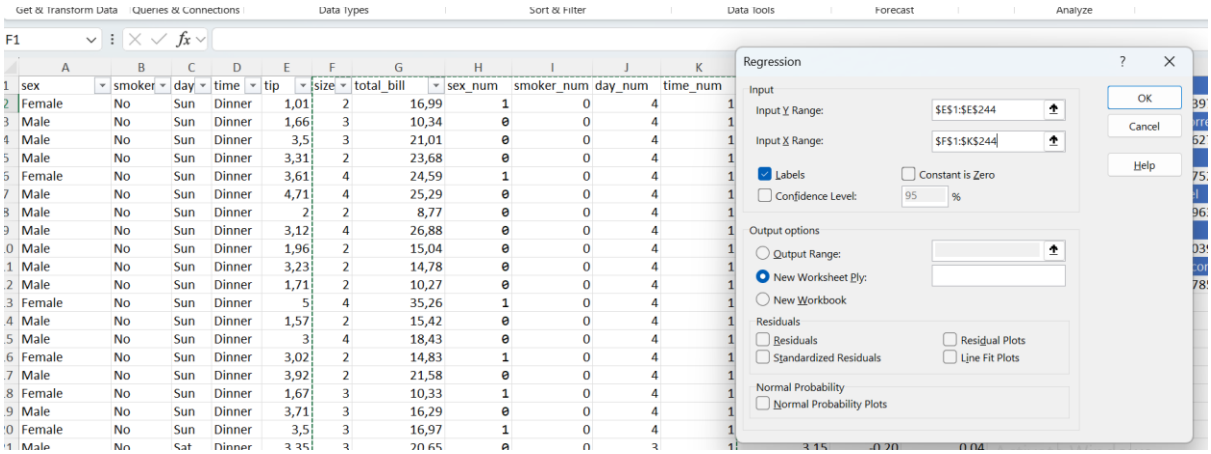


	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
er	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error			sex_correl		
	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	2,93			-0,085273975		
	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33			smoker_correl		
	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26	0,07			0,00976275		
	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01	0,00	MSE		day_correl		
	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18	0,03	1,02		0,131797526		
	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89	0,80	RMSE		time_correl		
	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09	0,01	1,01		0,11759639		
	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85	0,72			size_correl		
	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54	0,29			0,488400395		
	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75	0,57			total_bill_correl		
	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34	0,12			G244;E2:E244)		
	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21	0,04					
	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97	0,94					
	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17	0,03					
	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50	0,25					
	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80	0,64					

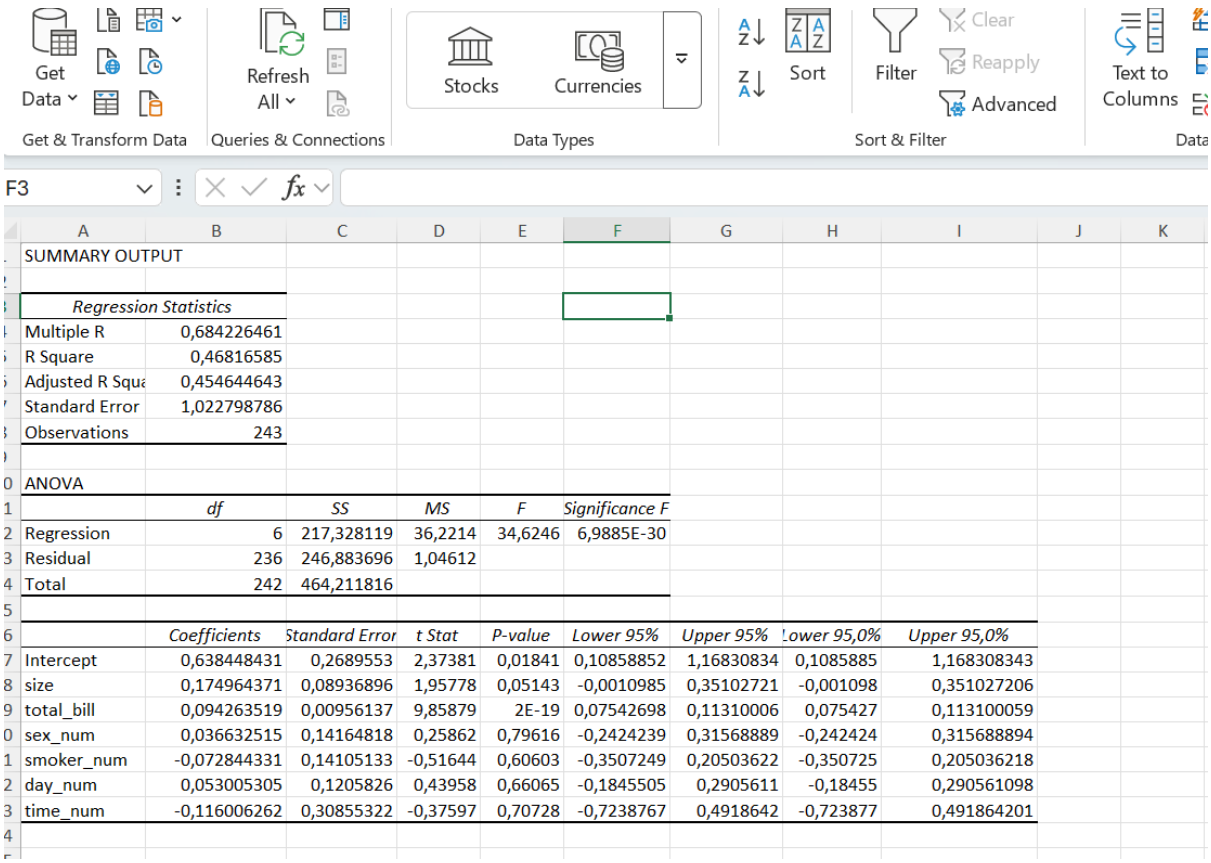


er	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error
	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71
	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57
	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26
	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01
	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18
	Sun	Dinner	4,71	4	25,29	0	0	4	1	3,82	-0,89
	Sun	Dinner	2	2	8,77	0	0	4	1	1,91	-0,09
	Sun	Dinner	3,12	4	26,88	0	0	4	1	3,97	0,85
	Sun	Dinner	1,96	2	15,04	0	0	4	1	2,50	0,54
	Sun	Dinner	3,23	2	14,78	0	0	4	1	2,48	-0,75
	Sun	Dinner	1,71	2	10,27	0	0	4	1	2,05	0,34
	Sun	Dinner	5	4	35,26	1	0	4	1	4,79	-0,21
	Sun	Dinner	1,57	2	15,42	0	0	4	1	2,54	0,97
	Sun	Dinner	3	4	18,43	0	0	4	1	3,17	0,17
	Sun	Dinner	3,02	2	14,83	1	0	4	1	2,52	-0,50
	Sun	Dinner	3,92	2	21,58	0	0	4	1	3,12	-0,80
	Sun	Dinner	1,67	3	10,33	1	0	4	1	2,27	0,60

	0,29
	0,57
	0,12
	0,04
	0,94
	0,03
	0,25
	0,64
	0,36



sex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num
Female	No	Sun	Dinner	1,01	2	16,99	1	0	4	1
Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1
Male	No	Sun	Dinner	3,5	3	21,01	0	0	4	1
Male	No	Sun	Dinner	3,31	2	23,68	0	0	4	1
Female	No	Sun	Dinner	3,61	4	24,59	1	0	4	1
Male	No	Sun	Dinner	4,71	4	25,29	0	0	4	1
Male	No	Sun	Dinner	2	2	8,77	0	0	4	1
Male	No	Sun	Dinner	3,12	4	26,88	0	0	4	1
Male	No	Sun	Dinner	1,96	2	15,04	0	0	4	1
Male	No	Sun	Dinner	3,23	2	14,78	0	0	4	1
Male	No	Sun	Dinner	1,71	2	10,27	0	0	4	1
Female	No	Sun	Dinner	5	4	35,26	1	0	4	1
Male	No	Sun	Dinner	1,57	2	15,42	0	0	4	1
Male	No	Sun	Dinner	3	4	18,43	0	0	4	1
Female	No	Sun	Dinner	3,02	2	14,83	1	0	4	1
Male	No	Sun	Dinner	3,92	2	21,58	0	0	4	1
Female	No	Sun	Dinner	1,67	3	10,33	1	0	4	1
Male	No	Sun	Dinner	3,71	3	16,29	0	0	4	1
Female	No	Sun	Dinner	3,5	3	16,97	1	0	4	1
Male	No	Sat	Dinner	3,35	3	20,65	0	0	3	1

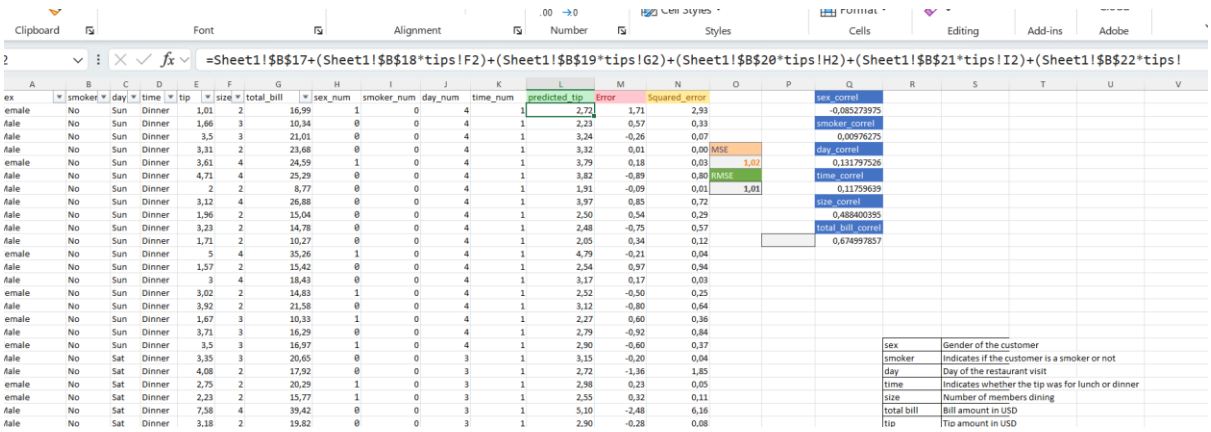


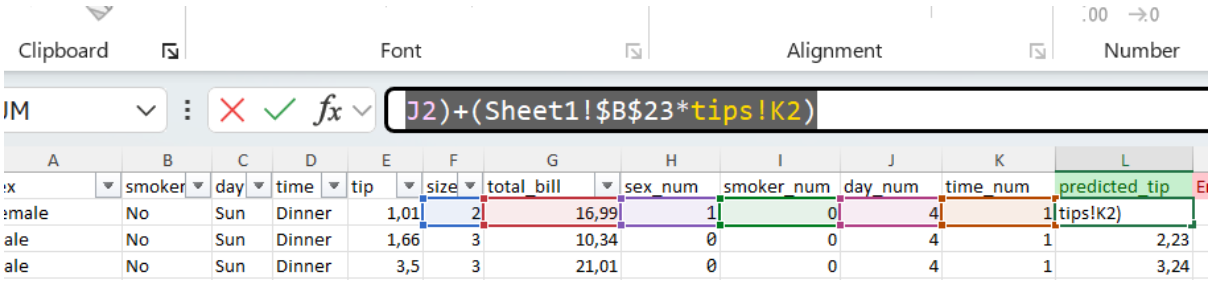
SUMMARY OUTPUT										
Regression Statistics										
Multiple R	0,684226461									
R Square	0,46816585									
Adjusted R Squa	0,454644643									
Standard Error	1,022798786									
Observations	243									
ANOVA										
	df	*SS*	*MS*	*F*	*Significance F*					
Regression	6	217,328119	36,2214	34,6246	6,9885E-30					
Residual	236	246,883696	1,04612							
Total	242	464,211816								
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95,0%*	*Upper 95,0%*		
Intercept	0,638448431	0,2689553	2,37381	0,01841	0,10858852	1,16830834	0,1085885	1,168308343		
size	0,174964371	0,08936896	1,95778	0,05143	-0,0010985	0,35102721	-0,001098	0,351027206		
total_bill	0,094263519	0,00956137	9,85879	2E-19	0,07542698	0,11310006	0,075427	0,113100059		
sex_num	0,036632515	0,14164818	0,25862	0,79616	-0,2424239	0,31568889	-0,242424	0,315688894		
smoker_num	-0,072844331	0,14105133	-0,51644	0,60603	-0,3507249	0,20503622	-0,350725	0,205036218		
day_num	0,053005305	0,1205826	0,43958	0,66065	-0,1845505	0,2905611	-0,18455	0,290561098		
time_num	-0,116006262	0,30855322	-0,37597	0,70728	-0,7238767	0,4918642	-0,723877	0,491864201		

Task6

Predicted Tip = Intercept + (Size_Coefficient * size) + (Total_Bill_Coefficient * total_bill) + (Sex_Coefficient * sex_num) + (Smoker_Coefficient * smoker_num) + (Day_Coefficient * day_num) + (Time_Coefficient * time_num)

Task7

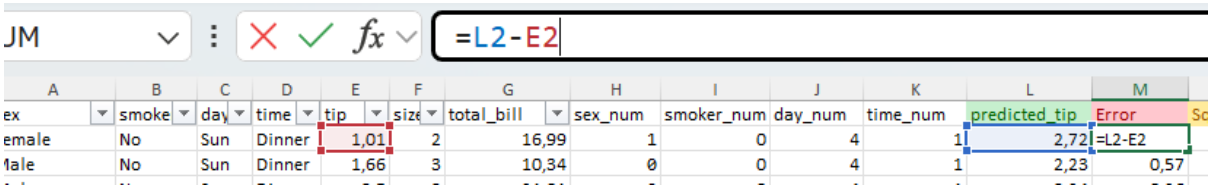




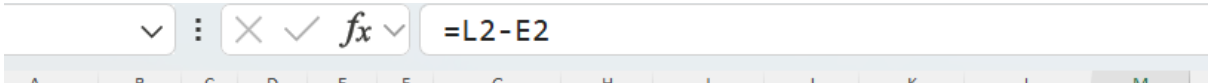
A	B	C	D	E	F	G	H	I	J	K	L
ex	smoker	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip
emale	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	tips!K2)
ale	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23
ale	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24

Task 8

error



A	B	C	D	E	F	G	H	I	J	K	L	M
ex	smoke	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error
emale	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	=L2-E2
Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57



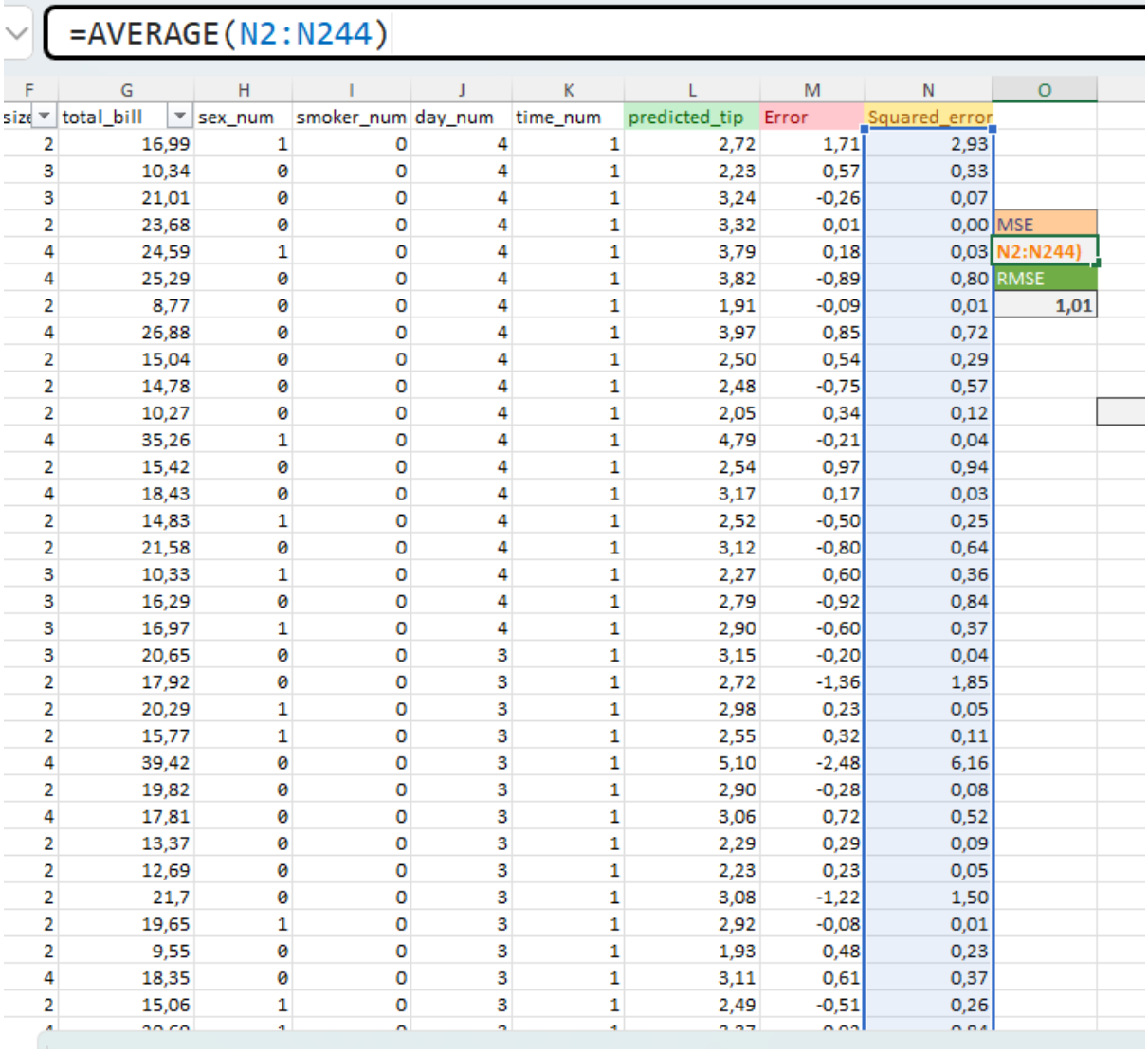
A	B	C	D	E	F	G	H	I	J	K	L	M
	smoke	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error
ale	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71
le	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57
le	No	Sun	Dinner	3,5	3	21,01	0	0	4	1	3,24	-0,26
le	No	Sun	Dinner	3,31	2	23,68	0	0	4	1	3,32	0,01
ale	No	Sun	Dinner	3,61	4	24,59	1	0	4	1	3,79	0,18

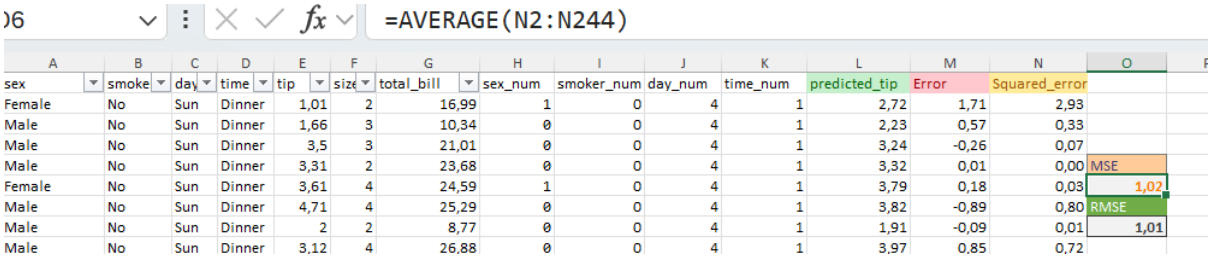
Squared error



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
sex	smoke	day	time	tip	size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error	
Female	No	Sun	Dinner	1,01	2	16,99	1	0	4	1	2,72	1,71	=M2^2	
Male	No	Sun	Dinner	1,66	3	10,34	0	0	4	1	2,23	0,57	0,33	
Male	No	Sun	Dinner	3.5	3	21.01	0	0	4	1	3.24	-0.26	0.07	

Mean Squared Error





Root mean squared error

=SQRT(O6)

F	G	H	I	J	K	L	M	N	O
size	total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error	
2	16,99	1	0	4	1	2,72	1,71	2,93	
3	10,34	0	0	4	1	2,23	0,57	0,33	
3	21,01	0	0	4	1	3,24	-0,26	0,07	
2	23,68	0	0	4	1	3,32	0,01	0,00	MSE
4	24,59	1	0	4	1	3,79	0,18	0,03	1,02
4	25,29	0	0	4	1	3,82	-0,89	0,80	RMSE
2	8,77	0	0	4	1	1,91	-0,09	0,01	=SQRT(O6)
4	26,88	0	0	4	1	3,97	0,85	0,72	

=SQRT(O6)

G	H	I	J	K	L	M	N	O	P
total_bill	sex_num	smoker_num	day_num	time_num	predicted_tip	Error	Squared_error		
16,99	1	0	4	1	2,72	1,71	2,93		
10,34	0	0	4	1	2,23	0,57	0,33		
21,01	0	0	4	1	3,24	-0,26	0,07		
23,68	0	0	4	1	3,32	0,01	0,00	MSE	
24,59	1	0	4	1	3,79	0,18	0,03	1,02	
25,29	0	0	4	1	3,82	-0,89	0,80	RMSE	
8,77	0	0	4	1	1,91	-0,09	0,01	1,01	
26,88	0	0	4	1	3,97	0,85	0,72		
15,04	0	0	4	1	2,50	0,54	0,29		